

Dev Kumar

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PROFESSIONAL SUMMARY

Senior Software Engineer with 8+ years shipping production RAG systems at scale, Neo4j knowledge graph platforms, and .NET/C# backend services on Azure. Owned end-to-end platforms serving millions of documents and tens of thousands of daily users — from ontology and schema design through ASP.NET Core APIs, Cypher queries, and observability. Applied engineering mindset, not research — proven record taking RAG and graph systems from architecture to production on small, ground-floor teams.

KEY SKILLS

RAG & LLM Systems: Retrieval-Augmented Generation (RAG), LLM integration, vector databases (Pinecone, Weaviate, Azure AI Search), embeddings, chunking strategies, reranking, OpenAI/Azure OpenAI APIs, LangChain

Knowledge Graphs: Neo4j, Cypher, APOC, GraphRAG, graph schema modeling, ontology design, entity extraction, relationship mapping, multi-hop traversals, Neo4j .NET driver, RDF/SPARQL

.NET & Languages: .NET 6/7/8, C#, ASP.NET Core, Entity Framework Core, LINQ, dependency injection, MediatR, Polly, SignalR, Minimal APIs, Python, C++, SQL, REST APIs, async/await

Azure Cloud: Azure OpenAI, Azure AI Search, Azure Functions, Azure App Service, Azure Cosmos DB, Blob Storage, Azure DevOps, Kubernetes (AKS)

Data & Systems: Document ingestion pipelines, ETL, PostgreSQL, MongoDB, Redis, Kafka, Docker, CI/CD, distributed systems, observability

PROFESSIONAL EXPERIENCE

Sprouts AI | *Sr. Software Engineer – RAG & Knowledge Systems* | Chicago, IL Mar 2025 – Present

- **Owned production RAG platform at scale — 5M+ docs, 50K+ daily queries** — architected Azure AI Search hybrid retrieval (vector + BM25) with semantic chunking, Azure OpenAI embeddings, and cross-encoder reranking; shipped behind feature flags with offline eval harness (MRR, nDCG, recall@k), boosting answer relevance by 42% over baseline.
- **Designed and owned company-wide Neo4j knowledge graph platform** — modeled domain ontology (20+ node labels, 40+ relationship types), authored Cypher schema constraints and indexes, and built GraphRAG hybrid that fuses graph traversals with vector retrieval, adopted as the system-of-record across 50+ internal teams.
- **Built production .NET 8 / ASP.NET Core backend services in C#** — designed minimal APIs with dependency injection, Polly retry/circuit-breaker policies, MediatR pipelines, Entity Framework Core over Cosmos DB, and Neo4j .NET driver integration, sustaining 10K+ daily RAG requests at sub-2s p95.
- **Owned scalability and maintainability of document analysis pipeline** — introduced chunking strategies, incremental indexing, and observability dashboards (OpenTelemetry + Azure Monitor), cutting reindex time by 70% and MTTR by 45%.

Resilience Inc | *Software Engineer – LLM Integrations & Backend* | Chicago, IL Aug 2023 – Mar 2025

- **Shipped production RAG Q&A system serving 50K+ monthly active users** — built document ingestion for 2M+ enterprise pages, hierarchical chunking with overlap tuning, vector similarity retrieval, and Azure OpenAI-backed answer synthesis with grounded citations, lifting answer accuracy by 38% and cutting support tickets by 25%.
- **Designed Neo4j knowledge graph schema with production Cypher queries** — modeled content ontology (courses, concepts, learners) with 15+ node types, authored parameterized Cypher queries and APOC procedures for multi-hop traversals, and built ETL from 100K+ documents, improving personalized recommendations by 40%.
- **Developed ASP.NET Core 6/7 microservices in C# deployed on Azure** — built REST APIs with JWT auth, middleware pipelines, EF Core migrations, and SignalR for real-time events; deployed via Azure App Service with App Configuration and Key Vault, sustaining 10K+ concurrent connections.
- **Owned end-to-end delivery of production LLM features** — collaborated with a small team on architecture, implementation, testing, and rollout; established CI/CD pipelines and observability that enabled zero-downtime deployments for mission-critical AI features.

Tata Steel | *Software Engineer (AI/ML) / IT Project Manager* Apr 2018 – Aug 2023

- **Architected and owned company-wide document analysis platform on Azure** — laid foundational .NET Core / C# platform adopted across the enterprise, ingesting millions of documents with ASP.NET Core APIs, embeddings-based semantic search, and metadata-driven retrieval, achieving 40% faster response times and 99.99% uptime.
- **Built knowledge graph foundation for enterprise learning platform** — designed graph schema linking courses, competencies, and employees; implemented graph-traversal queries that powered personalized learning recommendations across 20K+ users.
- **Shipped ASP.NET Core production services in C# integrating LLMs** — built prompt orchestration pipelines, embedding generation workers, and Azure-hosted vector retrieval components with EF Core and dependency injection, enabling AI-driven training personalization across the enterprise.
- **Established CI/CD and production observability practices** using Azure DevOps, Jenkins, and Git — automated model training, evaluation, and deployment workflows, reducing release cycle time by 30% and enabling reliable zero-downtime rollouts.

PROJECTS

LangChain – Chat with Search [Source](#) Python · LangChain · LLM · RAG

- Built a production RAG-style AI agent with LangChain, LLM-backed tool selection, and real-time retrieval across web search and Wikipedia — demonstrating document retrieval, LLM orchestration, and agentic reasoning foundational to enterprise RAG systems.

TradeSentinel [Source](#) | [Live Demo](#) Full-Stack · Production System

- Architected and shipped an end-to-end production platform with real-time data ingestion, structured storage, and graph-based relationship analysis — demonstrating applied software engineering from architecture through production deployment.

EDUCATION

Illinois Institute of Technology — M.S. in Computer Science

National Institute of Technology, Calicut — B.Tech in Computer Science

PUBLICATION

"Comparative study of movie recommendation system using feature engineering and improved error function," IEEE INCOFT 2021 | DOI:

10.1109/INCOFT55651.2022.10094480 | [Paper](#) | [Code](#) | [Certificate](#)